DO NOW — RECALL

# Networks, security and compression practice

Name: _____ Class: _____ Date: _____

**1. What does DNS resolve?**

**2. What is a router's role?**

**3. What is lossy compression?**

**4. $11010011_2$ AND $00001111_2$ as byte and hex?**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

**1. Model answer**

Domain name to an IP address.

**2. Model answer**

Forwards IP packets between networks.

**3. Model answer**

Discards information irreversibly.

**4. Model answer**

$00000011_2 = 03_{16}$.

**By the end, you can…**

• Distinguish DNS addressing, IP routing, TCP transport and TLS security: each serves a different purpose.

• RLE is lossless and suitable for runs; dictionary coding uses references. Lossy methods can discard detail.

• Hashing produces an integrity digest; symmetric and asymmetric encryption protect confidentiality using keys.

**Vocabulary — know these terms**

DNS — Maps names to IP addresses.

TLS — Authenticated secure transport.

RLE — Run-length encoding.

digest — Output of hash function.





# WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

**Guided 1 — original CHECK**

Explain and check

Explain the reason for the result.

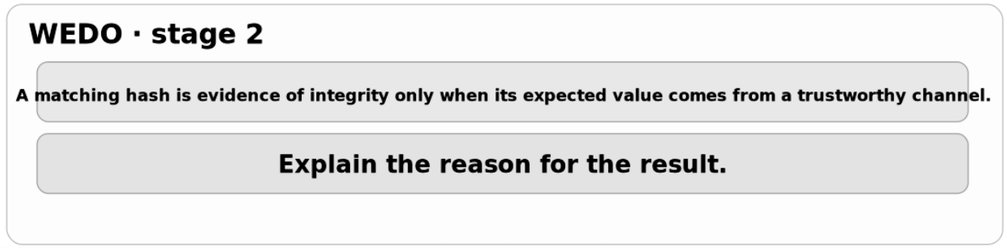
IMPROVE: Explain one correction to your own reasoning.

# We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Explain the limitation of a checksum displayed on the same compromised download website.**

A matching hash is evidence of integrity only when its expected value comes from a trustworthy channel.




WEDO · stage 2: A matching hash is evidence of integrity only when its expected value comes from a trustworthy channel.; Explain the reason for the result.





NEW LEARNING

# Read, annotate and explain

### Understand

Distinguish DNS addressing, IP routing, TCP transport and TLS security: each serves a different purpose.

### Apply

RLE is lossless and suitable for runs; dictionary coding uses references. Lossy methods can discard detail.

### Evaluate

Hashing produces an integrity digest; symmetric and asymmetric encryption protect confidentiality using keys.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**



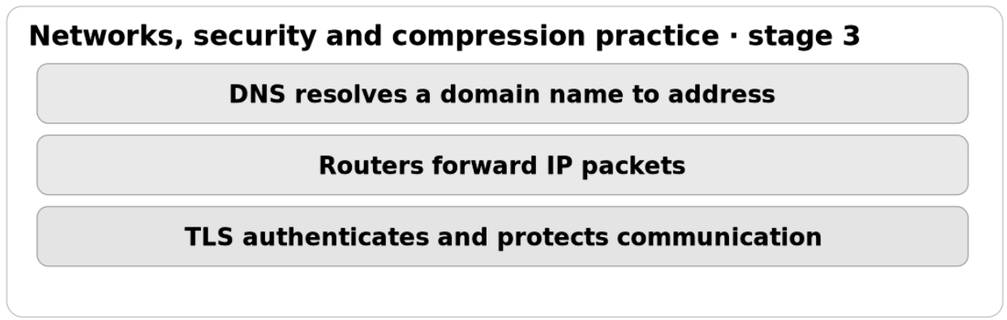

Networks, security and compression practice · stage 1: DNS resolves a domain name to address

**New Learning**




Networks, security and compression practice · stage 2: DNS resolves a domain name to address; Routers forward IP packets

**New Learning**




Networks, security and compression practice · stage 3: DNS resolves a domain name to address; Routers forward IP packets; TLS authenticates and protects communication





**EXIT TICKET / REFLECTION**

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**Hashes can be decrypted using the public key to recover the input.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

**Question 3**

Evaluate transmitting official exam results as a lossy image versus an authenticated, encrypted, losslessly stored file. [6]

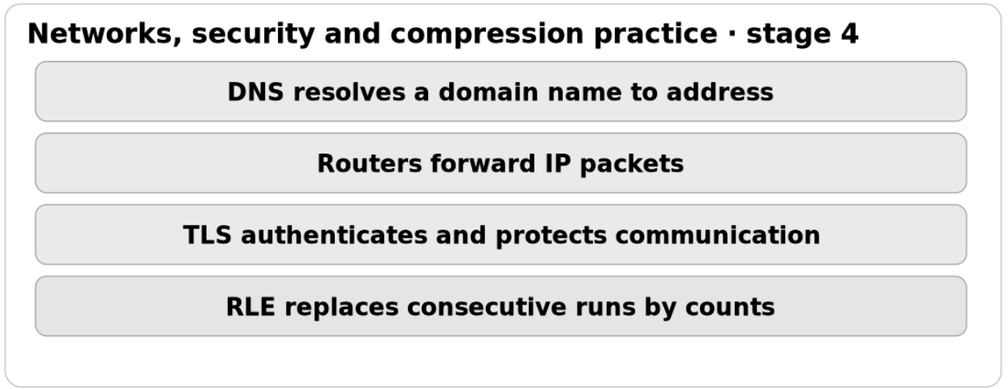




# New Learning · original lesson diagrams

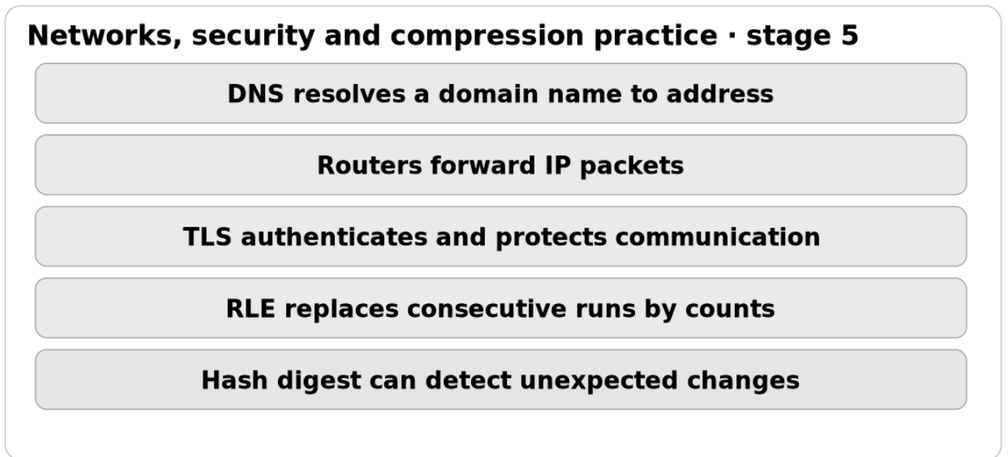
Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**




Networks, security and compression practice · stage 4: DNS resolves a domain name to address; Routers forward IP packets; TLS authenticates and protects communication; RLE replaces consecutive runs by counts

**New Learning**




Networks, security and compression practice · stage 5: DNS resolves a domain name to address; Routers forward IP packets; TLS authenticates and protects communication; RLE replaces consecutive runs by counts; Hash digest can detect unexpected changes

I DO — WORKED EXAMPLE

# Follow the worked method

**Encode AAABBCC with run-length encoding.**

**Predict first**

Count consecutive symbols, preserving the order of groups.

**Explain and check**

Explain the reason for the result.





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

**Question 1**

Describe DNS, router and TLS roles when accessing HTTPS website. [6]

**Question 2**

RLE-encode AAAAABBBCCDD and explain how decoding works. [6]

# WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

**Guided 1: Predict first A matching hash is evidence of integrity only when its expected value comes from a trustworthy channel.**

DISCUSS — Compare the representation or steps you used. Explain one decision and challenge one possible misconception.

IMPROVE — Write one precise correction or stronger explanation in your own answer.





# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Encode AAABBCC with run-length encoding.**

Count consecutive symbols, preserving the order of groups.

## IDO · stage 1

**Count consecutive symbols, preserving the order of groups.**

IDO · stage 1: Count consecutive symbols, preserving the order of groups.

**Encode AAABBCC with run-length encoding.**

Count consecutive symbols, preserving the order of groups.

## IDO · stage 2

**Count consecutive symbols, preserving the order of groups.**

**Explain the reason for the result.**

IDO · stage 2: Count consecutive symbols, preserving the order of groups.; Explain the reason for the result.

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve



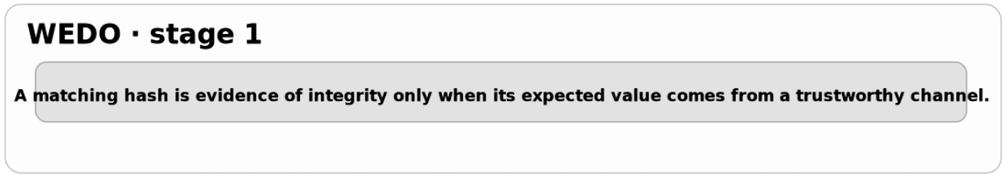


## We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Explain the limitation of a checksum displayed on the same compromised download website.**

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WEDO · stage 1: A matching hash is evidence of integrity only when its expected value comes from a trustworthy channel.